

CALCULUS I

Chapter 2: DERIVATIVES

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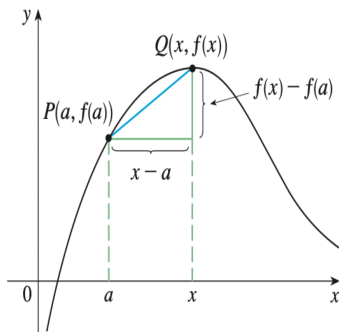
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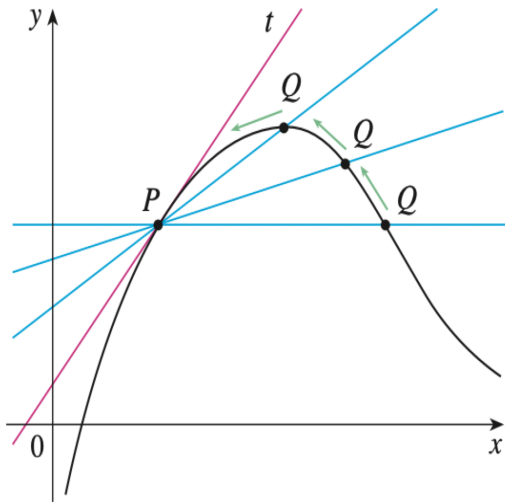
TANGENTS

If a curve C has equation $y = f(x)$ and we want to find the tangent line to C at the point $P(a, f(a))$, then we consider a nearby point $Q(x, f(x))$, where $x \neq a$, and compute the slope of the secant line PQ :

$$m_{PQ} = \frac{f(x) - f(a)}{x - a}$$



Then we let Q approach P along the curve C by letting x approach a . If m_{PQ} approaches a number m , then we define the **tangent** t to be the line through P with slope m .



Definition 1: The **tangent line** to the curve $y = f(x)$ at the point $P(a, f(a))$ is the line through P with slope

$$m = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

provided that this limit exists. In this case, the equation of the tangent line is

$$y - f(a) = m(x - a)$$

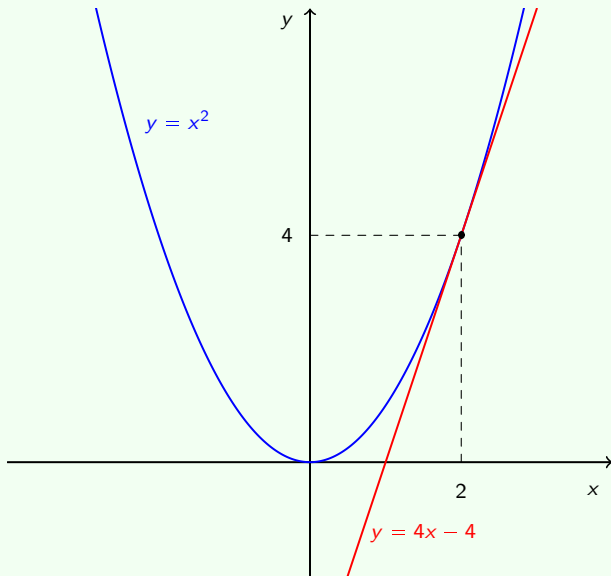
Example 1: Find an equation of the tangent line to the parabola $y = x^2$ at the point $P(2, 4)$.

Solution : We have $a = 2$ and $f(x) = x^2$, so the slope is

$$m = \lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} = \lim_{x \rightarrow 2} \frac{(x - 2)(x + 2)}{x - 2} = \lim_{x \rightarrow 2} (x + 2) = 4$$

The equation of the tangent line at $(2, 4)$ is

$$y - 4 = 4(x - 2) \Leftrightarrow y = 4x - 4$$



There is another expression for the slope of a tangent line that is sometimes easier to use. If $h = x - a$, then $x = a + h$, $x \rightarrow a \Rightarrow h \rightarrow 0$, and so the expression for the slope of the tangent line becomes

$$m = \lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h}$$

Example 2: Find an equation of the tangent line to the hyperbola $y = f(x) = 3/x$ at the point $(3, 1)$.

Solution : The slope of the tangent at $(3, 1)$ is

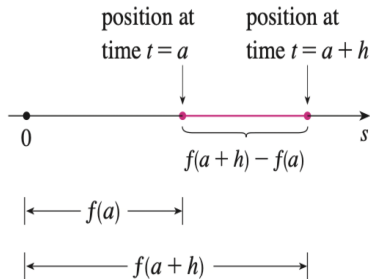
$$\begin{aligned} m &= \lim_{h \rightarrow 0} \frac{f(3+h) - f(3)}{h} = \lim_{h \rightarrow 0} \frac{\frac{3}{3+h} - 1}{h} \\ &= \lim_{h \rightarrow 0} \frac{-1}{3+h} = -\frac{1}{3} \end{aligned}$$

Therefore an equation of the tangent at the point $(3, 1)$ is

$$y - 1 = -\frac{1}{3}(x - 3) \Leftrightarrow x + 3y - 6 = 0$$

VELOCITIES

Suppose an object moves along a straight line according to an equation of motion $s = f(t)$, where s is the **displacement** (directed distance) of the object from the origin at time t . The function f that describes the motion is called the **position function** of the object. In the time interval from $t = a$ to $t = a + h$ the change in position is $f(a + h) - f(a)$.



The **average velocity** over this time interval is

$$\text{average velocity} = \frac{\text{displacement}}{\text{time}} = \frac{f(a+h) - f(a)}{h}$$

If now we let h approach 0, then we come to the definition of velocity.

Definition 2: The **velocity** (or **instantaneous velocity**) at the time $t = a$ of an object, that moves along a straight line with the displacement $s = f(t)$, is defined as follows:

$$v(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Example 3: Suppose that a ball is dropped from the upper observation deck of the tower, 450m above the ground.

- (a) What is the velocity of the ball after 5 seconds?
- (b) How fast is the ball traveling when it hits the ground?

Solution : From Galileo's law, the distance fallen after t seconds is $s = f(t) = 4.9t^2(m)$. We start by finding the velocity at a general time $t = a$.

$$\begin{aligned} v(a) &= \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = \lim_{h \rightarrow 0} \frac{4.9(a+h)^2 - 4.9a^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{4.9(2ah + h^2)}{h} = \lim_{h \rightarrow 0} (9.8a + 4.9h) = 9.8a(m/s) \end{aligned}$$

- (a) The velocity after 5s is: $v(5) = 9.8 \times 5 = 49(m/s)$
- (b) Since the observation deck is 450 m above the ground, the ball will hit the ground at the time t_1 when $s(t_1) = 450$, that is,

$$4.9t_1^2 = 450 \Rightarrow t_1 = \sqrt{\frac{450}{4.9}} \approx 9.6s$$

The velocity of the ball as it hits the ground is therefore

$$v(t_1) = 9.8t_1 = 9.8\sqrt{\frac{450}{4.9}} \approx 94(m/s)$$

DERIVATIVES

Definition 3: The derivative of a function f at a number a , denoted by $f'(a)$, is

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

if this limit exists. (Note that $f'(a)$ is read "f prime of a"). If we write $x = a + h$, then we have $h = x - a$ and h approaches 0 if and only if x approaches a . Therefore an equivalent way of stating the definition of the derivative is

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

Example 4: Find the derivative of the function

$$f(x) = x^3 + 2x + 3$$

at the number a .

Solution : From the definition we have

$$\begin{aligned} f'(a) &= \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} \\ &= \lim_{h \rightarrow 0} \frac{[(a+h)^3 + 2(a+h) + 3] - [a^3 + 2a + 3]}{h} \\ &= \lim_{h \rightarrow 0} \frac{3a^2h + 3ah^2 + h^3 + 2h}{h} \\ &= \lim_{h \rightarrow 0} [(3a^2 + 2) + 3ah + h^2] = 3a^2 + 2 \end{aligned}$$

NOTE:

The tangent line to $y = f(x)$ at $(a, f(a))$ is the line through $(a, f(a))$ whose slope is equal to $f'(a)$. Therefore the equation of the tangent line is

$$y - f(a) = f'(a)(x - a)$$

Example 5: Find an equation of the tangent line to the curve $y = x^3 + 2x + 3$ at the point $P(1, 6)$.

Solution : From the previous example, we have

$$f'(1) = 3 \cdot 1^2 + 2 = 5$$

The equation of the tangent line is

$$y - 6 = 5(x - 1) \iff y = 5x + 1$$

Definition 4: The **left-hand derivative** of f at the point a , denoted by $f'(a^-)$ or $f'_-(a)$, is

$$f'(a^-) = \lim_{x \rightarrow a^-} \frac{f(x) - f(a)}{x - a}$$

The **right-hand derivative** of f at the point a , denoted by $f'(a^+)$ or $f'_+(a)$, is

$$f'(a^+) = \lim_{x \rightarrow a^+} \frac{f(x) - f(a)}{x - a}$$

Theorem 1: *The function f has the derivative at a if and only if f has the left-hand and right-hand derivatives at a and they are equal.*

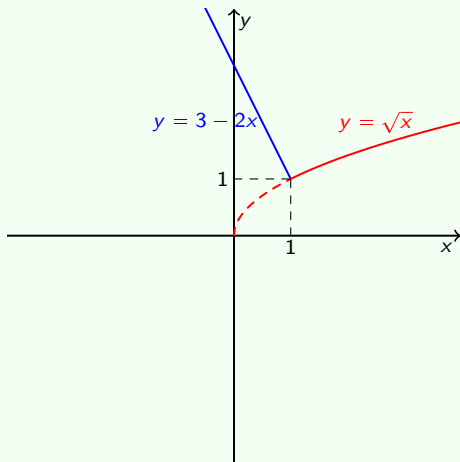
Example 6: Given $f(x) = \begin{cases} 3 - 2x, & \text{if } x < 1 \\ \sqrt{x}, & \text{if } x \geq 1 \end{cases}$. Find the one-sided derivatives of f at 1.

Solution : We have $f(1) = 1$ and

$$f'(1^-) = \lim_{x \rightarrow 1^-} \frac{f(x) - f(1)}{x - 1} = \lim_{x \rightarrow 1^-} \frac{(3 - 2x) - 1}{x - 1} = -2$$

$$\begin{aligned} f'(1^+) &= \lim_{x \rightarrow 1^+} \frac{f(x) - f(1)}{x - 1} = \lim_{x \rightarrow 1^+} \frac{\sqrt{x} - 1}{(\sqrt{x})^2 - 1} \\ &= \lim_{x \rightarrow 1^+} \frac{\sqrt{x} - 1}{(\sqrt{x} - 1)(\sqrt{x} + 1)} = \lim_{x \rightarrow 1^+} \frac{1}{\sqrt{x} + 1} = \frac{1}{2} \end{aligned}$$

We see that $f'(1^-) = -2 \neq f'(1^+) = \frac{1}{2}$, thus the function has no derivative at 1 (although f is continuous at 1).



RATES OF CHANGE

Suppose y is a quantity that depends on another quantity x . Thus y is a function of x and we write $y = f(x)$. If x changes from x_1 to x_2 , then the change in x (also called the **increment** of x) is

$$\Delta x = x_2 - x_1$$

and the corresponding change in y is

$$\Delta y = f(x_2) - f(x_1)$$

The difference quotient

$$\frac{\Delta y}{\Delta x} = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

is called the **average rate of change** of y with respect to x over the interval $[x_1, x_2]$.

Now by letting x_2 approach x_1 and therefore letting Δx approach 0, we have the limit of these average rates of change is called the **rate of change** of y with respect to x at $x = x_1$:

$$\text{rate of change} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{x_2 \rightarrow x_1} \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

We recognize this limit as being the derivative $f'(x_1)$. We now have an interpretation

Theorem 2 (Interpretation of the Derivative): *The derivative $f'(a)$ is the rate of change of $y = f(x)$ with respect to x when $x = a$.*

THE DERIVATIVE AS A FUNCTION

We considered the derivative of a function f at a fixed number a :

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Now we let the number a vary and if we replace a by a variable x , then we obtain

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

So we can regard $f'(x)$ as a new function of x , called the **derivative function**, or shortly, **derivative** of f .

Example 7: Find the derivative of $f(x) = \sqrt{x}$.

Solution : We have

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} \\ &= \lim_{h \rightarrow 0} \frac{(\sqrt{x+h} - \sqrt{x})(\sqrt{x+h} + \sqrt{x})}{h(\sqrt{x+h} + \sqrt{x})} \\ &= \lim_{h \rightarrow 0} \frac{(x+h) - x}{h(\sqrt{x+h} + \sqrt{x})} = \lim_{h \rightarrow 0} \frac{1}{\sqrt{x+h} + \sqrt{x}} \\ &= \frac{1}{\sqrt{x} + \sqrt{x}} = \frac{1}{2\sqrt{x}} \end{aligned}$$

We see that the domain of $f(x)$ is $[0, +\infty)$ and the domain of $f'(x)$ is $(0, +\infty)$.

If we use the traditional notation $y = f(x)$ to indicate that the independent variable is x and the dependent variable is y , then some common alternative notations for the derivative are as follows:

$$f'(x) = y' = \frac{dy}{dx} = \frac{df}{dx} = \frac{d}{dx}f(x)$$

If we want to indicate the value of a derivative $\frac{dy}{dx}$ at a specific number a , we use the notation

$$f'(a) = \left. \frac{dy}{dx} \right|_{x=a}$$

DERIVATIVES OF THE FUNDAMENTAL FUNCTIONS

We have the following formulas:

$$(1) \quad (x^\alpha)' = \alpha x^{\alpha-1}, \quad (\alpha \in \mathbb{R})$$

$$(2) \quad (a^x)' = a^x \ln a, \quad (a > 0, a \neq 1) \quad \text{and} \quad (e^x)' = e^x$$

$$(3) \quad (\log_a x)' = \frac{1}{x \ln a}, \quad (a > 0, a \neq 1) \quad \text{and} \quad (\ln x)' = \frac{1}{x}$$

$$(4) \quad (\sin x)' = \cos x, \quad (\cos x)' = -\sin x$$

$$(5) \quad (\tan x)' = \frac{1}{\cos^2 x}, \quad (\cot x)' = -\frac{1}{\sin^2 x}$$

$$(6) \quad (\arcsin x)' = -(\arccos x)' = \frac{1}{\sqrt{1-x^2}}$$

$$(7) \quad (\arctan x)' = \frac{1}{1+x^2}$$

$$(8) \quad (\sinh x)' = \cosh x, \quad (\cosh x)' = \sinh x$$

$$(9) \quad (\tanh x)' = \frac{1}{\cosh^2 x}, \quad (\coth x)' = -\frac{1}{\sinh^2 x}$$

DIFFERENTIABLE FUNCTIONS

Definition 5: A function f is **differentiable** at a if $f'(a)$ exists. It is differentiable on an open interval (a, b) (a, b may be infinite) if it is differentiable at every number in the interval.

Example 8: Where is the function $f(x) = |x|$ differentiable?

Solution : If $x > 0$ then $|x| = x$ and we can choose h small enough that $x + h > 0$ and $|x + h| = x + h$. Therefore, for $x > 0$, we have

$$f'(x) = \lim_{h \rightarrow 0} \frac{|x + h| - |x|}{h} = \lim_{h \rightarrow 0} \frac{x + h - x}{h} = 1$$

so f is differentiable for all $x > 0$.

Similarly, for $x < 0$, we have $|x| = -x$ and h can be chosen small enough that $x + h < 0$ and $|x + h| = -(x + h)$. Therefore, for $x < 0$, we have

$$f'(x) = \lim_{h \rightarrow 0} \frac{|x + h| - |x|}{h} = \lim_{h \rightarrow 0} \frac{-(x + h) - (-x)}{h} = -1$$

so f is differentiable for all $x < 0$.

For $x = 0$ we have $\lim_{h \rightarrow 0} \frac{|0 + h| - |0|}{h} = \lim_{h \rightarrow 0} \frac{|h|}{h}$ and

$$\lim_{h \rightarrow 0^-} \frac{|h|}{h} = \lim_{h \rightarrow 0^-} \frac{-h}{h} = -1, \quad \lim_{h \rightarrow 0^+} \frac{|h|}{h} = \lim_{h \rightarrow 0^+} \frac{h}{h} = 1$$

Since these limits are different, $f'(0)$ does not exist. Thus f is differentiable for all x except 0.

Theorem 3: *If f is differentiable at a , then it is continuous at a .*

NOTE

The converse of this theorem is false; that is, there are functions that are continuous but not differentiable. For instance, the function $f(x) = |x|$ is continuous at 0, but it is not differentiable at 0.

HIGHER DERIVATIVES

If f is a differentiable function, then its derivative f' is also a function, so f' may have a derivative of its own, denoted by $(f')' = f''$. This new function f'' is called the **second derivative** of f because it is the derivative of the derivative of f . We can write the second derivative of $y = f(x)$ as

$$f''(x) = y'' = \frac{d^2 y}{dx^2} = \frac{d^2 f}{dx^2} = \frac{d^2}{dx^2} f(x)$$

The **third derivative** is the derivative of the second derivative: $f''' = (f'')'$. We denote

$$f'''(x) = y''' = \frac{d^3 y}{dx^3} = \frac{d^3 f}{dx^3} = \frac{d^3}{dx^3} f(x)$$

The process can be continued. In general, the n^{th} derivative of f is denoted by $f^{(n)}(x)$, ($n > 3$) and is obtained from f by differentiating n times. If $f = f(x)$, we write

$$f^{(n)}(x) = y^{(n)} = \frac{d^n y}{dx^n} = \frac{d^n f}{dx^n} = \frac{d^n}{dx^n} f(x)$$

n^{th} derivatives of some fundamental functions:

$$(1) (e^x)^{(n)} = e^x$$

$$(2) (\sin x)^{(n)} = \sin\left(x + \frac{n\pi}{2}\right), (\cos x)^{(n)} = \cos\left(x + \frac{n\pi}{2}\right)$$

$$(3) [(1+x)^\alpha]^{(n)} = \alpha(\alpha-1)\dots(\alpha-n+1)(1+x)^{\alpha-n}$$

$$(4) \left[\frac{1}{1+x}\right]^{(n)} = \frac{(-1)^n n!}{(1+x)^{n+1}}$$

$$(5) [\ln(1+x)]^{(n)} = \frac{(-1)^{n-1} (n-1)!}{(1+x)^n}, \quad (n \geq 1)$$

RULES OF DIFFERENTIATION

Definition 6 (The Constant Multiple Rule):

$$[Cf(x)]' = Cf'(x), \quad (C : \text{constant})$$

Definition 7 (The Sum Rule):

$$[f(x) \pm g(x)]' = f'(x) \pm g'(x)$$

NOTE:

The Sum Rule can be extended to the sum of any number of functions. For instance,

$$(f + g - h)' = f' + g' - h'$$

Example 9: Find the points on the curve $y = x^4 - 6x^2 + 4$ where the tangent line is horizontal.

Solution : Horizontal tangents occur where the derivative is zero. We have

$$y' = (x^4 - 6x^2 + 4)' = 4x^3 - 12x = 4x(x^2 - 3)$$

Thus $y' = 0$ if $x = 0$ or $x = \pm\sqrt{3}$. So the given curve has horizontal tangents when $x = 0$ or $x = \pm\sqrt{3}$. The corresponding points are $(0, 4)$, $(\sqrt{3}, -5)$, and $(-\sqrt{3}, -5)$.

Definition 8 (The Product Rule):

$$[f(x) \cdot g(x)]' = f'(x) \cdot g(x) + f(x) \cdot g'(x)$$

Example 10: If $h(x) = xg(x)$ and it is known that $g(2) = 7$, $g'(2) = 5$, find $h'(2)$.

We have

$$h'(x) = (x)'g(x) + x(g(x))' = g(x) + xg'(x)$$

Therefore $h'(2) = g(2) + 2g'(2) = 7 + 2 \times 5 = 17$.

NOTE:

The Product Rule can be extended to the product of any number of functions. For instance,

$$(fgh)' = f'gh + fg'h + fgh'$$

And in general,

$$\left(\prod_{k=1}^n f_k \right)' = \sum_{k=1}^n f_1 f_2 \dots f_k' \dots f_n$$

Definition 9 (The Quotation Rule):

$$\left[\frac{f(x)}{g(x)} \right]' = \frac{f'(x) \cdot g(x) - f(x) \cdot g'(x)}{[g(x)]^2}$$

Example 11:

$$\left(\frac{x^2}{x^3 + 1} \right)' = \frac{(x^2)'(x^3 + 1) - x^2(x^3 + 1)'}{(x^3 + 1)^2} = \frac{2x - x^4}{(x^3 + 1)^2}$$

$$\left(\frac{\sin x}{x} \right)' = \frac{(\sin x)' \cdot x - \sin x \cdot (x)'}{x^2} = \frac{x \cos x - \sin x}{x^2}$$

Example 12: Find equations of the tangent line and normal line to the curve $y = \frac{\sqrt{x}}{1+x^2}$ at the point $(1, \frac{1}{2})$.

Solution : We have

$$\begin{aligned}y' &= \frac{(\sqrt{x})'(1+x^2) - \sqrt{x}(1+x^2)'}{(1+x^2)^2} \\&= \frac{\frac{1}{2\sqrt{x}}(1+x^2) - \sqrt{x}(2x)}{(1+x^2)^2} \\&= \frac{(1+x^2) - 4x^2}{2\sqrt{x}(1+x^2)^2} = \frac{1-3x^2}{2\sqrt{x}(1+x^2)^2}\end{aligned}$$

The slope of the tangent line at $(1, \frac{1}{2})$ is $y'(1) = -\frac{1}{4}$ and the slope of the normal line is the negative reciprocal of $-\frac{1}{4}$, that is 4. So the equations are:

The tangent line: $y - \frac{1}{2} = -\frac{1}{4}(x - 1) \Leftrightarrow 4y + x - 3 = 0$

The normal line: $y - \frac{1}{2} = 4(x - 1) \Leftrightarrow y - 4x + \frac{7}{2} = 0$

Definition 10 (The Chain Rule): If $u(x)$ is differentiable at x and $f(u)$ is differentiable at $u(x)$, then the composite function $F(x) = f(u(x))$ is differentiable at x and

$$F'(x) = f'(u(x)) \cdot u'(x) \quad \text{or} \quad \frac{dF}{dx} = \frac{dF}{du} \cdot \frac{du}{dx}$$

Example 13: Find the derivative of $y = \sqrt{x^2 + 1}$.

Solution : Denote $u(x) = x^2 + 1$ and $f(u) = \sqrt{u}$. We have

$$y(x) = f(u(x)), \quad f'(u) = \frac{1}{2\sqrt{u}}, \quad u' = 2x, \quad \text{and}$$

$$y' = f'(u) \cdot u'(x) = \frac{1}{2\sqrt{u}} \cdot 2x = \frac{1}{2\sqrt{x^2 + 1}} \cdot 2x = \frac{x}{\sqrt{x^2 + 1}}$$

From the Chain Rule, we have the following useful formulas:

$$(1) [u^\alpha(x)]' = \alpha \cdot u^{\alpha-1}(x) \cdot u'(x)$$

$$(2) [\sqrt{u(x)}]' = \frac{u'(x)}{2\sqrt{u(x)}}$$

$$(3) [a^{u(x)}]' = u'(x) \cdot a^{u(x)} \cdot \ln a, \quad (a > 0, a \neq 1)$$

$$(4) [e^{u(x)}]' = u'(x) \cdot e^{u(x)}$$

$$(5) [\ln u(x)]' = \frac{u'(x)}{u(x)}$$

IMPLICIT DIFFERENTIATION

The functions that we have met so far can be described by expressing one variable explicitly in terms of another variable, for example, $y = \sqrt{x^2 + 1}$ or $y = x \cos x$ or, in general, $y = f(x)$. Some functions, however, are defined implicitly by a relation between x and y such as

$$x^2 + y^2 = 25 \quad \text{or} \quad x^3 + y^3 - 6xy = 0$$

In order to find the derivative of y , we can use the method of **implicit differentiation**. This consists of differentiating both sides of the equation with respect to x and then solving the resulting equation for y' . In the examples and exercises of this section it is always assumed that the given equation determines y implicitly as a differentiable function of x so that the method of implicit differentiation can be applied.

Example 14:

- (a) If $x^2 + y^2 = 25$, find $y'(x)$.
- (b) Find an equation of the tangent to the circle $x^2 + y^2 = 25$ at the point $(3, 4)$.

Solution :

- (a) Differentiate both sides of the equation $x^2 + y^2 = 25$ with respect to x :

$$(x^2 + y^2)'_x = (25)'_x \Rightarrow 2x + 2yy' = 0 \Rightarrow y' = -\frac{x}{y}$$

- (b) At the point $(3, 4)$, we have $y' = -\frac{3}{4}$. Therefore, the equation of the tangent line is

$$y - 4 = -\frac{3}{4}(x - 3) \Leftrightarrow 3x + 4y = 25$$

Example 15:

- (a) If $x^3 + y^3 = 6xy$, find $y'(x)$.
- (b) Find an equation of the tangent to the folium of Descartes $x^3 + y^3 = 6xy$ at the point $(3, 3)$.
- (c) At what point in the first quadrant is the tangent line horizontal?

Solution :

- (a) Differentiate both sides of the equation $x^3 + y^3 = 6xy$ with respect to x :

$$3x^2 + 3y^2y' = 6y + 6xy' \Rightarrow y' = \frac{2y - x^2}{y^2 - 2x}$$

- (b) At the point $(3, 3)$, $y' = -1$, and the equation of the tangent is $y - 3 = -1(x - 3) \Leftrightarrow x + y = 6$.
- (c) The tangent line is horizontal if $y' = 0$, that is $2y - x^2 = 0$ (provided that $y^2 - 2x \neq 0$). Substituting $y = \frac{1}{2}x^2$ in the equation of the curve, we get

$$x^3 + \left(\frac{1}{2}x^2\right)^3 = 6x \left(\frac{1}{2}x^2\right) \Leftrightarrow x^6 = 16x^3$$

Since $x > 0$ in the first quadrant, we get

$$x^3 = 16 \Rightarrow x = 2\sqrt[3]{2} \quad \text{then} \quad y = \frac{1}{2}x^2 = 2\sqrt[3]{4}$$

Thus the tangent is horizontal at $(2\sqrt[3]{2}, 2\sqrt[3]{4})$.

Example 16: Find y'' if $x^4 + y^4 = 1$.

Solution : Differentiating the equation implicitly with respect to x , we get

$$4x^3 + 4y^3y' = 0 \implies y' = -\frac{x^3}{y^3}$$

We now differentiate this expression with respect to x again:

$$\begin{aligned} y'' &= \left(-\frac{x^3}{y^3} \right)' = -\frac{(x^3)'y^3 - x^3(y^3)'}{y^6} \\ &= -\frac{3x^2y^3 - 3x^3y^2y'}{y^6} = -\frac{3x^2y - 3x^3\left(-\frac{x^3}{y^3}\right)}{y^4} \\ &= -\frac{3x^2y^4 + 3x^6}{y^7} = -\frac{3x^2(y^4 + x^4)}{y^7} = -\frac{3x^2}{y^7} \end{aligned}$$

RATES OF CHANGE IN SCIENCES

PHYSICS

If $s = f(t)$ is the position function of a particle that is moving in a straight line, then $\frac{\Delta s}{\Delta t}$ represents the **average velocity** over a time period Δt , and $v(t) = \frac{ds}{dt}$ represents the **velocity** (the rate of change of displacement with respect to time). The rate of change of velocity with respect to time is **acceleration**: $a(t) = v'(t) = s''(t)$.

Example 17: The position of a particle is given by the equation

$$s = f(t) = t^3 - 6t^2 + 9t$$

where t is measured in seconds and s in meters.

- (a) Find the velocity at time t .
- (b) What is the velocity after 2s? After 4s?
- (c) When is the particle at rest?
- (d) When is the particle moving forward (that is, in the positive direction)?
- (e) Draw a diagram to represent the motion of the particle.
- (f) Find the total distance traveled by the particle during the first five seconds.
- (g) Find the acceleration at time t and after 4s.
- (h) Graph the position, velocity, and acceleration functions for $0 \leq t \leq 5$.
- (i) When is the particle speeding up, slowing down?

Solution :

- (a) The velocity function is the derivative of the position function.

$$s = f(t) = t^3 - 6t^2 + 9t \Rightarrow v(t) = 3t^2 - 12t + 9$$

- (b) The velocities after 2s and 4s are

$$v(2) = 3(2)^2 - 12(2) + 9 = -3 \text{ and}$$

$$v(4) = 3(4)^2 - 12(4) + 9 = 9$$

- (c) The particle is at rest when $v(t) = 0$, that is,

$$3t^2 - 12t + 9 = 3(t - 1)(t - 3) = 0$$

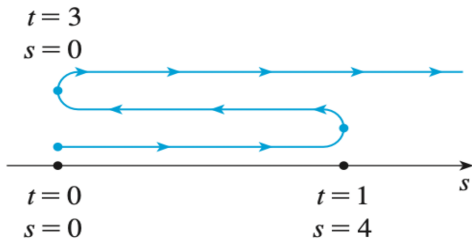
and this is true when $t = 1$ or $t = 3$. Thus the particle is at rest after 1s and after 3s.

(d) The particle moves in the positive direction when $v(t) > 0$, that is,

$$3t^2 - 12t + 9 = 3(t - 1)(t - 3) > 0$$

This inequality is true when both factors are positive $t > 3$ or when both factors are negative $t < 1$. Thus the particle moves in the positive direction in the time intervals $t < 1$ and $t > 3$. It moves backward (in the negative direction) when $1 < t < 3$.

(e) We have the motion of the particle along a line (the s -axis).



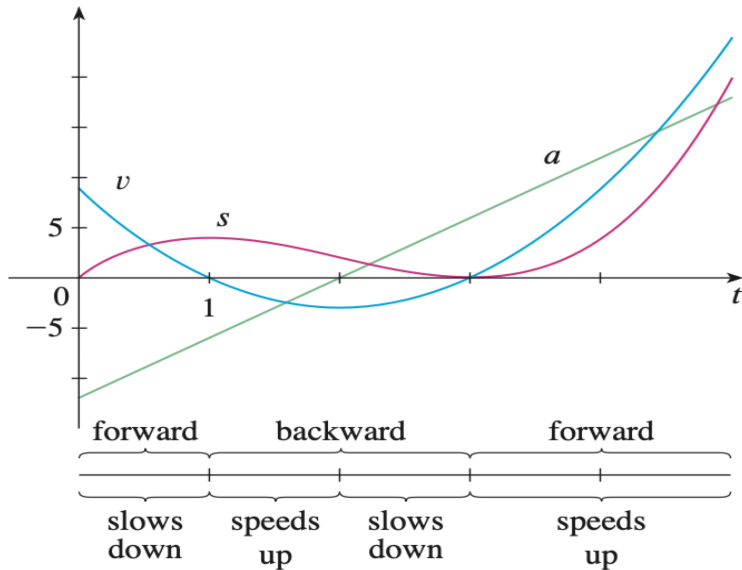
- (f) We need to calculate the distances traveled during the time intervals $[0, 1]$, $[1, 3]$, and $[3, 5]$ separately, and the total distance is

$$\begin{aligned} T &= |f(1) - f(0)| + |f(3) - f(1)| + |f(5) - f(3)| \\ &= 4 + 4 + 20 = 28 \end{aligned}$$

- (g) The acceleration is the derivative of the velocity function:

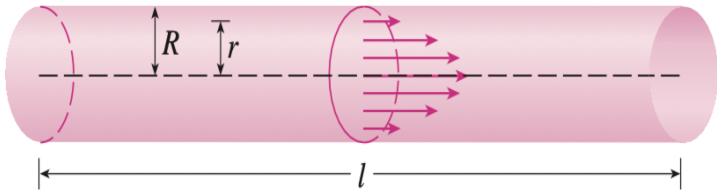
$$a(t) = v'(t) = 6t - 12 \quad \text{and} \quad a(4) = 12(m/s^2)$$

- (h) The next figure shows the graphs of s , v , and a
- (i) The particle speeds up when the velocity is positive and increasing (v and a are both positive) and also when the velocity is negative and decreasing (v and a are both negative). In other words, the particle speeds up when the velocity and acceleration have the same sign. (The particle is pushed in the same direction it is moving.) From the figure we see that this happens when $1 < t < 2$ and when $t > 3$. The particle slows down when v and a have opposite signs, that is, when $0 < t < 1$ and when $2 < t < 3$.



BIOLOGY

When we consider the flow of blood through a blood vessel, such as a vein or artery, we can model the shape of the blood vessel by a cylindrical tube with radius R and length l as illustrated in the following figure.



Because of friction at the walls of the tube, the velocity v of the blood is greatest along the central axis of the tube and decreases as the distance r from the axis increases until v becomes 0 at the wall. The relationship between v and r is given by the law of laminar flow:

$$v = \frac{P}{4\eta l}(R^2 - r^2)$$

where η is the viscosity of the blood and P is the pressure difference between the ends of the tube. If P and l are constant, then v is a function of r with domain $[0, R]$.

The average rate of change of the velocity as we move from $r = r_1$ outward to $r = r_2$ is given by

$$\frac{\Delta v}{\Delta r} = \frac{v(r_2) - v(r_1)}{r_2 - r_1}$$

and if we let $\Delta r \rightarrow 0$, we obtain the **velocity gradient**, that is, the instantaneous rate of change of velocity with respect to r :

$$\text{velocity gradient} = \lim_{\Delta r \rightarrow 0} \frac{\Delta v}{\Delta r} = \frac{dv}{dr} = -\frac{Pr}{2\eta l}$$

For one of the smaller human arteries we can take $\eta = 0.027$, $R = 0.008\text{cm}$, $l = 2\text{cm}$, and $P = 4000 \text{ dynes}/\text{cm}^2$, which gives

$$v = \frac{4000}{4(0.027)^2}(0.008^2 - r^2) = 1.85 \times 10^4(6.4 \times 10^{-5} - r^2)$$

At $r = 0.002\text{cm}$ the blood is flowing at a speed of

$$v = 1.85 \times 10^4(6.4 \times 10^{-5} - 0.002^2) \approx 1.11\text{cm/s}$$

and the velocity gradient at that point is

$$\left. \frac{dv}{dr} \right|_{r=0.002} = -\frac{4000(0.002)}{2(0.027)^2} \approx -74 \text{ (1/s)}$$

ECONOMICS

Suppose $C(x)$ is the total cost that a company incurs in producing x units of a certain commodity. The function C is called a **cost function**. If the number of items produced is increased from x_1 to x_2 , then the additional cost is $\Delta C = C(x_2) - C(x_1)$, and the average rate of change of the cost is

$$\frac{\Delta C}{\Delta x} = \frac{C(x_2) - C(x_1)}{x_2 - x_1} = \frac{C(x_1 + \Delta x) - C(x_1)}{\Delta x}$$

The limit of this quantity as $\Delta x \rightarrow 0$, that is, the rate of change of cost with respect to the number of items produced, is called the **marginal cost** by economists:

$$\text{marginal cost} = \lim_{\Delta x \rightarrow 0} \frac{\Delta C}{\Delta x} = \frac{dC}{dx}$$

Since x often takes on only integer values ($x = n$), it may not make literal sense to let x approach 0, but we can always replace $C(x)$ by a smooth approximating function.

Taking $\Delta x = 1$ and n large (so that Δx is small compared to n), we have

$$C'(n) \approx C(n+1) - C(n)$$

Thus the marginal cost of producing n units is approximately equal to the cost of producing one more unit [the $(n + 1)^{th}$ unit].

It is often appropriate to represent a total cost function by a polynomial

$$C(x) = A + Bx + Cx^2 + Dx^3$$

where A represents the overhead cost (rent, heat, maintenance) and the other terms represent the cost of raw materials, labor, and so on. (The cost of raw materials may be proportional to x , but labor costs might depend partly on higher powers of x because of overtime costs and inefficiencies involved in large-scale operations.)

Example 18: Suppose a company has estimated that the cost (in dollars) of producing x items is

$$C(x) = 10,000 + 5x + 0.01x^2$$

Then the marginal cost function is

$$C'(x) = 5 + 0.02x$$

The marginal cost at the production level of 500 items is

$$C'(500) = 5 + 0.02 \times 500 = \$15 \text{ per item}$$

This gives the rate at which costs are increasing with respect to the production level when $x = 500$ and predicts the cost of the 501st item. The actual cost of producing the 501st item is

$$\begin{aligned} C(501) - C(500) &= (10,000 + 5(501) + 0.01(501)^2) \\ &\quad - (10,000 + 5(500) + 0.01(500)^2) \\ &= \$15.01 \end{aligned}$$

Notice that $C'(500) \approx C(501) - C(500)$.

RELATED RATES

THE COMMON IDEA

In a related rates problem the idea is to compute the rate of change of one quantity in terms of the rate of change of another quantity (which may be more easily measured). The procedure is to find an equation that relates the two quantities and then use the Chain Rule to differentiate both sides with respect to time.

Example 19: Air is being pumped into a spherical balloon so that its volume increases at a rate of $100 \text{ cm}^3/\text{s}$. How fast is the radius of the balloon increasing when the diameter is 50 cm?

Solution : We start by identifying two things:

the given: the rate of increase of the volume of air is
 $100\text{cm}^3/\text{s}$

the unknown: the rate of increase of the radius when the
diameter is 50cm

In order to express these quantities mathematically, we introduce some suggestive notation: Let V be the volume of the balloon and let r be its radius. The key thing to remember is that rates of change are derivatives. In this problem, the volume and the radius are both functions of the time t .

The rate of increase of the volume with respect to time is the derivative $\frac{dV}{dt}$, and the rate of increase of the radius is $\frac{dr}{dt}$. We can therefore restate the given and the unknown as follows:

the given: $\frac{dV}{dt} = 100\text{cm}^3/\text{s}$

the unknown: $\frac{dr}{dt}$ when $r = 25\text{cm}$

In order to connect dV/dt and dr/dt , we first relate V and r by the formula for the volume of a sphere:

$$V = \frac{4}{3}\pi r^3$$

In order to use the given information, we differentiate each side of this equation with respect to t . To differentiate the right side, we need to use the Chain Rule:

$$\frac{dV}{dt} = \frac{dV}{dr} \cdot \frac{dr}{dt} = 4\pi r^2 \frac{dr}{dt} \Rightarrow \frac{dr}{dt} = \frac{1}{4\pi r^2} \frac{dV}{dt}$$

If we put $r = 25$ and $dV/dt = 100$ in this equation, we obtain

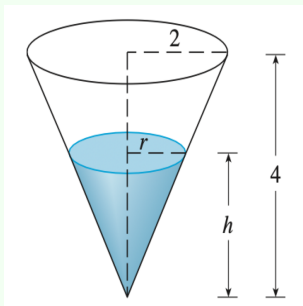
$$\frac{dr}{dt} = \frac{1}{4\pi(25)^2} \cdot 100 = \frac{1}{25\pi}$$

The radius of the balloon is increasing at the rate of

$$\frac{1}{25\pi} \approx 0.0127 \text{ cm/s}$$

Example 20:

A water tank has the shape of an inverted circular cone with base radius $2m$ and height $4m$. If water is being pumped into the tank at a rate of $2m^3/min$, find the rate at which the water level is rising when the water is $3m$ deep.



Solution : We first sketch the cone and label it as in the figure. Let V , r , and h be the volume of the water, the radius of the surface, and the height of the water at time t , where t is measured in minutes.

We are given that $dV/dt = 2\text{m}^3/\text{min}$ and we are asked to find dh/dt when h is 3m . The quantities V and h are related by the equation

$$V = \frac{1}{3}\pi r^2 h$$

but it is very useful to express V as a function of h alone. In order to eliminate r , we use the similar triangles in the figure to write

$$\frac{r}{h} = \frac{2}{4} \Rightarrow r = \frac{h}{2}$$

and the expression for V becomes

$$V = \frac{1}{3}\pi \left(\frac{h}{2}\right)^2 h = \frac{\pi}{12}h^3$$

Now we can differentiate each side with respect to t :

$$\frac{dV}{dt} = \frac{\pi}{4}h^2 \cdot \frac{dh}{dt} \Rightarrow \frac{dh}{dt} = \frac{4}{\pi h^2} \cdot \frac{dV}{dt}$$

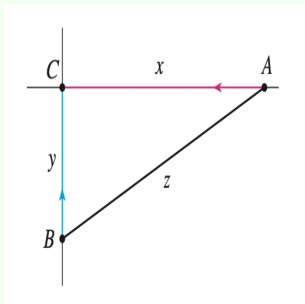
Substituting $h = 3m$ and $dV/dt = 2m^3/min$, we have

$$\frac{dh}{dt} = \frac{4}{\pi(3)^2} \cdot 2 = \frac{8}{9\pi}$$

The water level is rising at a rate of $\frac{8}{9\pi} \approx 0.28m/min$.

Example 21:

Car A is traveling west at 50mi/h and car B is traveling north at 60mi/h . Both are headed for the intersection of the two roads. At what rate are the cars approaching each other when car A is 0.3mi and car B is 0.4mi from the intersection?



Solution : Suppose C is the intersection of the roads. At a given time t , let x be the distance from car A to C, let y be the distance from car B to C, and let z be the distance between the cars, where x , y , and z are measured in miles. We are given that $dx/dt = -50\text{mi/h}$ and $dy/dt = -60\text{mi/h}$. (The derivatives are negative because x and y are decreasing.) We are asked to find dz/dt . The equation that relates x , y , and z is given by the Pythagorean Theorem:

$$z^2 = x^2 + y^2$$

Differentiating each side with respect to t , we have

$$2z \frac{dz}{dt} = 2x \frac{dx}{dt} + 2y \frac{dy}{dt} \quad \Rightarrow \quad \frac{dz}{dt} = \frac{1}{z} \left(x \frac{dx}{dt} + y \frac{dy}{dt} \right)$$

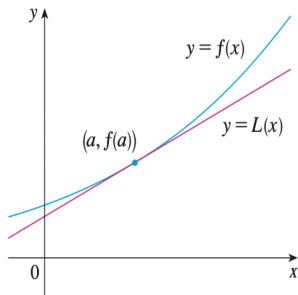
When $x = 0.3mi$ and $y = 0.4mi$, we have $z = 0.5mi$, so

$$\frac{dz}{dt} = \frac{1}{0.5} [0.3(-50) + 0.4(-60)] = -78mi/h$$

The cars are approaching each other at a rate of $78mi/h$.

LINEAR APPROXIMATIONS AND DIFFERENTIALS

The problem is that it might be easy to calculate a value $f(a)$ of a function, but difficult (or even impossible) to compute nearby values of f . So we settle for the easily computed values of the linear function L whose graph is the tangent line of f at $(a, f(a))$.



In other words, we use the tangent line at $(a, f(a))$ as an approximation to the curve $y = f(x)$ when x is near a . An equation of this tangent line is $y = f(a) + f'(a)(x - a)$ and the approximation

$$f(x) \approx f(a) + f'(a)(x - a)$$

is called the **linear approximation** or **tangent line approximation** of f at a .

The linear function whose graph is this tangent line, that is,

$$L(x) = f(a) + f'(a)(x - a)$$

is called the **linearization** of f at a .

Example 22: Find the linearization of the function $f(x) = \sqrt{x}$ at $a = 4$ and use it to approximate the numbers $\sqrt{3.98}$ and $\sqrt{4.05}$.

Solution : The derivative of $f(x) = x^{1/2}$ is $f'(x) = \frac{1}{2\sqrt{x}}$, and so we have $f(4) = 2$, $f'(4) = \frac{1}{4}$. Thus the linearization of f is $L(x) = 2 + \frac{1}{4}(x - 4) = 1 + \frac{x}{4}$. The corresponding linear approximation is $\sqrt{x} \approx 1 + \frac{x}{4}$ when x is near 4. In particular, we have

$$\sqrt{3.98} \approx 1 + \frac{3.98}{4} = 1.9950 \quad (\text{the true value is } 1.994993734)$$

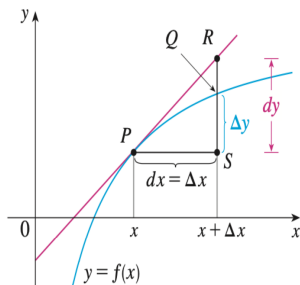
$$\sqrt{4.05} \approx 1 + \frac{4.05}{4} = 2.0125 \quad (\text{the true value is } 2.01246118)$$

Definition 11 (Differentials): The ideas behind linear approximations are sometimes formulated in the terminology and notation of *differentials*. If $y = f(x)$, where f is a differentiable function, then the differential dx is an independent variable; that is, dx can be given the value of any real number. The differential dy is then defined in terms of dx by the equation

$$dy = f'(x)dx$$

So dy is a dependent variable; it depends on the values of x and dx . If dx is given a specific value and x is taken to be some specific number in the domain of f , then the numerical value of dy is determined.

The geometric meaning of differentials is shown in the figure. Let $P(x, f(x))$ and $Q(x + \Delta x, f(x + \Delta x))$ be points on the graph of f and let $dx = \Delta x$. The corresponding change in y is $\Delta y = f(x + \Delta x) - f(x)$.



The slope of the tangent line PR is the derivative $f'(x)$. Thus the directed distance from S to R is $f'(x)dx = dy$. Therefore dy represents the amount that the tangent line rises or falls, whereas Δy represents the amount that the curve $y = f(x)$ rises or falls when x changes by an amount dx .

Example 23: Compare the values of dy and Δy if $f(x) = x^3 - 2x^2 + 3x - 4$ and x changes from 2 to 2.05.

Solution : We have

$$f(2) = 2, f(2.05) = 2.360125 \Rightarrow \Delta y = 0.360125$$

In general,

$$dy = f'(x)dx = (3x^2 - 4x + 3)dx$$

When $x = 2$ and $dx = \Delta x = 0.05$, this becomes

$$dy = [3(2)^3 - 4(2) + 3]0.05 = 0.35$$

Example 24: The radius of a sphere was measured and found to be 21 cm with a possible error in measurement of at most 0.05 cm. What is the maximum error in using this value of the radius to compute the volume of the sphere?

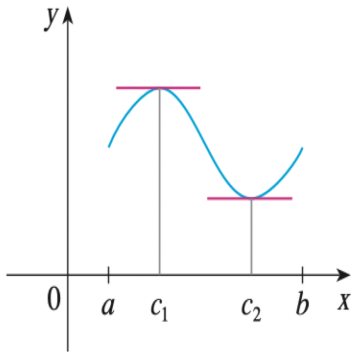
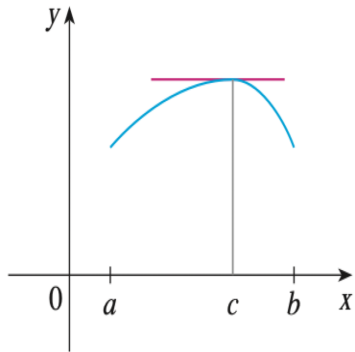
Solution : If the radius of the sphere is r , then its volume is $V = \frac{4}{3}\pi r^3$. If the error in the measured value of r is denoted by $dr = \Delta r$, then the corresponding error in the calculated value of V is ΔV , which can be approximated by the differential $dV = 4\pi r^2 dr$. When $r = 21$ and $dr = 0.05$, this becomes $dV = 4\pi(21)^2(0.05) \approx 277$. The maximum error in the calculated volume is about 277 cm^3 .

THE MEAN VALUE THEOREM

Theorem 4 (Rolle's Theorem): *Let f be a function that satisfies the following hypotheses:*

- (a) *f is continuous on the closed interval $[a, b]$.*
- (b) *f is differentiable on the open interval (a, b) .*
- (c) *$f(a) = f(b)$*

Then there is a number c in (a, b) such that $f'(c) = 0$.



Example 25: Prove that the equation $x^3 + x - 1 = 0$ has exactly one real root.

Solution : First we use the Intermediate Value Theorem to show that a root exists. Let $f(x) = x^3 + x - 1$. Then $f(0) = -1 < 0$, $f(1) = 1 > 0$. Since f is a polynomial, it is continuous, so the Intermediate Value Theorem states that there is a number c between 0 and 1 such that $f(c) = 0$. Thus the given equation has a root.

To show that the equation has no other real root, we use Rolle's Theorem and argue by contradiction. Suppose that it had two roots a and b . Then

$$f(a) = 0 = f(b)$$

and, since f is a polynomial, it is differentiable on (a, b) and continuous on $[a, b]$. Thus, by Rolle's Theorem, there is a number c between a and b such that $f'(c) = 0$. But

$$f'(x) = 3x^2 + 1 \geq 1$$

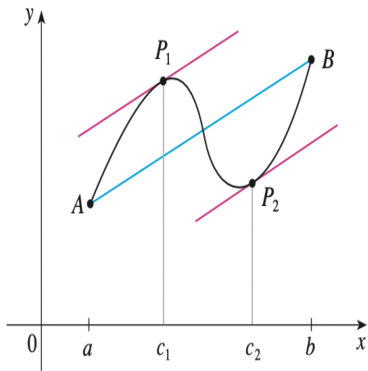
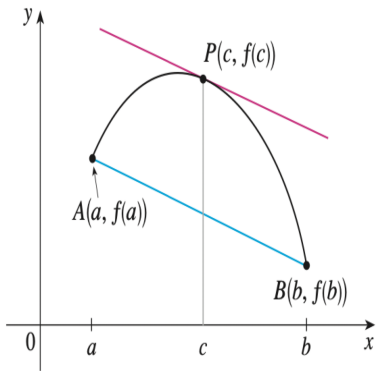
for all x so $f'(x)$ can never be 0. This gives a contradiction. Therefore the equation can't have two real roots.

Theorem 5 (The Mean Value Theorem): *Let f be a function that satisfies the following hypotheses:*

- (a) *f is continuous on the closed interval $[a, b]$.*
- (b) *f is differentiable on the open interval (a, b) .*

Then there is a number c in (a, b) such that

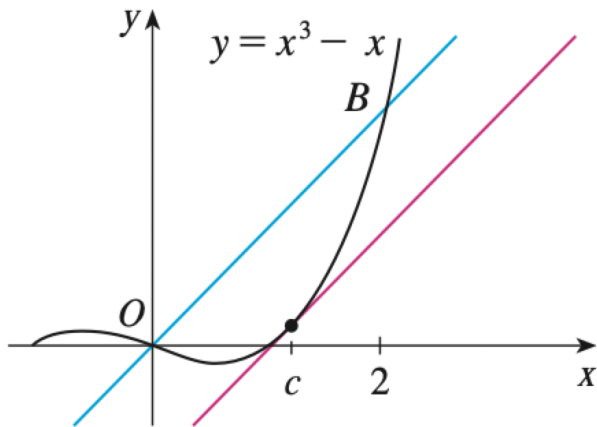
$$f'(c) = \frac{f(b) - f(a)}{b - a} \quad \Leftrightarrow \quad f(b) - f(a) = f'(c)(b - a)$$



Example 26: To illustrate the Mean Value Theorem with a specific function, let's consider $f(x) = x^3 - x$, $a = 0$, $b = 2$. Since f is a polynomial, it is continuous and differentiable for all x , so it is certainly continuous on $[0, 2]$ and differentiable on $(0, 2)$. Therefore, by the Mean Value Theorem, there is a number c in $(0, 2)$ such that

$$\begin{aligned}f(2) - f(0) &= f'(c)(2 - 0) \Leftrightarrow 6 - 0 = (3c^2 - 1)(2 - 0) \\&\Leftrightarrow 6 = 6c^2 - 2\end{aligned}$$

which gives $c = \pm \frac{2}{\sqrt{3}}$. But c must lie in $(0, 2)$, so $c = \frac{2}{\sqrt{3}}$. The next figure illustrates this calculation: The tangent line at this value of c is parallel to the secant line OB .



Example 27: Suppose that $f(0) = -3$ and $f'(x) \leq 5$ for all values of x . How large can $f(2)$ possibly be?

Solution : We are given that f is differentiable (and therefore continuous) everywhere. In particular, we can apply the Mean Value Theorem on the interval $[0, 2]$. There exists a number c such that

$$f(2) - f(0) = f'(c)(2 - 0) \Rightarrow f(2) = f(0) + 2f'(c) = -3 + 2f'(c)$$

Since $f'(x) \leq 5$ for all x , we have

$$f(2) = -3 + 2f'(c) \leq -3 + 2 \times 5 = 7$$

The largest possible value for $f(2)$ is 7.

Theorem 6: *If $f'(x) = 0$ for all x in an interval (a, b) , then f is constant on (a, b) .*

Corollary 1: If $f'(x) = g'(x)$ for all x in an interval (a, b) , then $f - g$ is constant on (a, b) ; that is, $f(x) = g(x) + C$ where C is a constant.

L'HOSPITAL'S RULE

Theorem 7 (L'Hospital's Rule): Suppose f and g are differentiable and $g(x) \neq 0$ on an open interval I that contains a (except possibly at a). Suppose that

$$\lim_{x \rightarrow a} f(x) = 0 \quad \text{and} \quad \lim_{x \rightarrow a} g(x) = 0$$

or that

$$\lim_{x \rightarrow a} f(x) = \pm\infty \quad \text{and} \quad \lim_{x \rightarrow a} g(x) = \pm\infty$$

In other words, we have an indeterminate form of type $\frac{0}{0}$ or $\frac{\infty}{\infty}$. Then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

if the limit on the right side exists (or is ∞ or $-\infty$).

NOTES:

- (1) L'Hospital's Rule says that the limit of a quotient of functions is equal to the limit of the quotient of their derivatives, provided that the given conditions are satisfied. It is especially important to verify the conditions regarding the limits of f and g before using L'Hospital's Rule.
- (2) L'Hospital's Rule is also valid for one-sided limits and for limits at infinity or negative infinity; that is, $x \rightarrow a$ can be replaced by any of the symbols $x \rightarrow a^-$, $x \rightarrow a^+$, $x \rightarrow +\infty$, or $x \rightarrow -\infty$.
- (3) L'Hospital's Rule can be used for more than one time.

Example 28:

$$\lim_{x \rightarrow 0^+} x \ln x = \lim_{x \rightarrow 0^+} \frac{\ln x}{1/x} = \lim_{x \rightarrow 0^+} \frac{1/x}{-1/x^2} = \lim_{x \rightarrow 0^+} (-x) = 0$$

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \lim_{x \rightarrow 0} \frac{\sin x}{2x} = \frac{1}{2} \lim_{x \rightarrow 0} \frac{\sin x}{x} = \frac{1}{2}$$

Example 29: Find $\lim_{x \rightarrow \infty} \frac{x + \sin x}{x - \cos x}$. We have $(x + \sin x)' = 1 + \cos x$, $(x - \cos x)' = 1 + \sin x$, $\lim_{x \rightarrow \infty} \frac{1 + \cos x}{1 + \sin x}$ does not exist, and we conclude that $\lim_{x \rightarrow \infty} \frac{x + \sin x}{x - \cos x}$ does not exist.

This is wrong !. The correct answer must be:

$$\lim_{x \rightarrow \infty} \frac{x + \sin x}{x - \cos x} = \lim_{x \rightarrow \infty} \frac{1 + \frac{\sin x}{x}}{1 - \frac{\cos x}{x}} = 1$$

TAYLOR'S AND MACLAURIN'S FORMULAS

We remember the n^{th} derivatives ($n \in \mathbb{N}$) of some fundamental functions:

$$(1) \quad (e^x)^{(n)} = e^x$$

$$(2) \quad (\sin x)^{(n)} = \sin \left(x + \frac{n\pi}{2} \right)$$

$$(3) \quad (\cos x)^{(n)} = \cos \left(x + \frac{n\pi}{2} \right)$$

$$(4) \quad [(1+x)^\alpha]^{(n)} = \alpha(\alpha-1)\dots(\alpha-n+1)(1+x)^{\alpha-n}$$

$$(5) \quad \left[\frac{1}{1+x} \right]^{(n)} = \frac{(-1)^n n!}{(1+x)^{n+1}}$$

$$(6) \quad [\ln(1+x)]^{(n)} = \frac{(-1)^{n-1} (n-1)!}{(1+x)^n}, \quad (n \geq 1)$$

Definition 12: If $f(x)$ has the n^{th} derivatives on an open interval I that contains a number a , then we have the following Taylor's formula

$$f(x) = f(a) + \frac{f'(a)}{1!}(x-a) + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n + R_n(x)$$

If $a = 0$, we have MacLaurin's formula:

$$f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \cdots + \frac{f^{(n)}(0)}{n!}x^n + R_n(x)$$

We consider MacLaurin's formula for some functions.

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \cdots + \frac{x^n}{n!} + R_n(x)$$

$$\sin x = x - \frac{x^3}{3!} + \cdots + (-1)^k \frac{x^{2k+1}}{(2k+1)!} + R_n(x)$$

$$\cos x = 1 - \frac{x^2}{2!} + \cdots + (-1)^k \frac{x^{2k}}{(2k)!} + R_n(x)$$

$$\frac{1}{1+x} = 1 - x + x^2 - x^3 + \cdots + (-1)^n x^n + R_n(x)$$

Example 30: Find MacLaurin's formula of $f(x) = \frac{2}{x^2 - 3x + 2}$ to 3rd order.

Solution : We have

$$f(x) = \frac{2}{x^2 - 3x + 2} = \frac{2}{(x-1)(x-2)} = 2 \left[\frac{1}{x-2} - \frac{1}{x-1} \right]$$

$$f'(x) = 2 \left[\frac{-1}{(x-2)^2} + \frac{1}{(x-1)^2} \right]$$

$$f''(x) = 2 \left[\frac{2}{(x-2)^3} - \frac{2}{(x-1)^3} \right]$$

$$f'''(x) = 2 \left[\frac{-6}{(x-2)^4} + \frac{6}{(x-1)^4} \right]$$

Then

$$f(0) = 1, f'(0) = \frac{3}{2}, f''(0) = \frac{7}{2}, f'''(0) = \frac{45}{4}$$

Thus

$$\begin{aligned}\frac{2}{x^2 - 3x + 2} &= f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + R_3(x) \\ &= 1 + \frac{3/2}{1}x + \frac{7/2}{2}x^2 + \frac{45/4}{6}x^3 + R_3(x) \\ &= 1 + \frac{3}{2}x + \frac{7}{4}x^2 + \frac{15}{8}x^3 + R_3(x)\end{aligned}$$

Example 31: Find Taylor's formula of $f(x) = x^2 \ln x$ to 3rd order at $x_0 = 1$ and use this formula to approximate $f(1.1)$.

Solution : We have $f(x) = x^2 \ln x \Rightarrow f(1) = 0$, $f'(x) = 2x \ln x + x \Rightarrow f'(1) = 1$, $f''(x) = 2 \ln x + 3 \Rightarrow f''(1) = 3$, $f'''(x) = \frac{2}{x} \Rightarrow f'''(1) = 2$. Thus

$$\begin{aligned} f(x) &= f(1) + \frac{f'(1)}{1!}(x-1) + \frac{f''(1)}{2!}(x-1)^2 \\ &\quad + \frac{f'''(1)}{3!}(x-1)^3 + R_3(x) = 0 + \frac{1}{1!}(x-1) \\ &\quad + \frac{3}{2!}(x-1)^2 + \frac{2}{3!}(x-1)^3 + R_3(x) \\ &= (x-1) + \frac{3}{2}(x-1)^2 + \frac{1}{3}(x-1)^3 + R_3(x) \end{aligned}$$

In order to approximate $f(1.1)$ we use the following formula (we omit the infinitesimal $R_3(x)$): for all x in the neighbourhood of 1

$$f(x) \approx (x - 1) + \frac{3}{2}(x - 1)^2 + \frac{1}{3}(x - 1)^3$$

We obtain

$$\begin{aligned} f(1.1) &\approx (1.1 - 1) + \frac{3}{2}(1.1 - 1)^2 + \frac{1}{3}(1.1 - 1)^3 \\ &= 0.\underline{115333333} \end{aligned}$$

The actual value is $f(1.1) = 1.1^2 \ln 1.1 = 0.\underline{1153253176}$

MAXIMUM AND MINIMUM VALUES

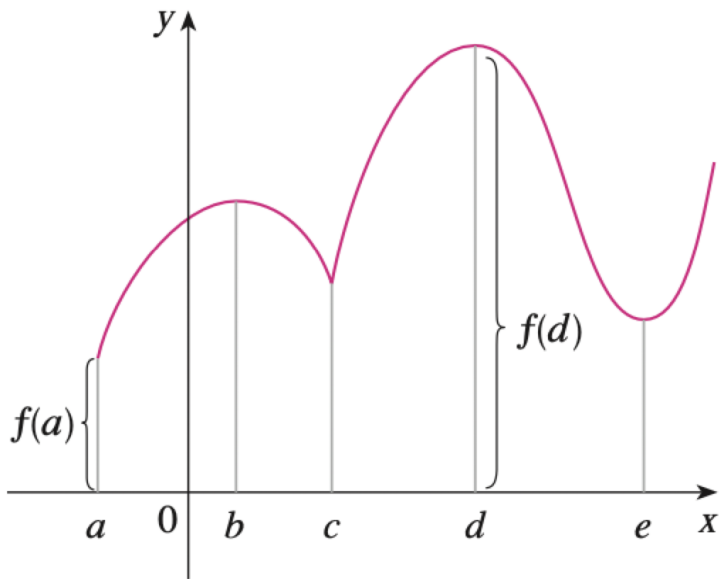
Definition 13: Let c be a number in the domain D of a function f . Then $f(c)$ is the

- **absolute maximum** value of f on D if $f(x) \leq f(c)$ for all x in D .
- **absolute minimum** value of f on D if $f(x) \geq f(c)$ for all x in D .

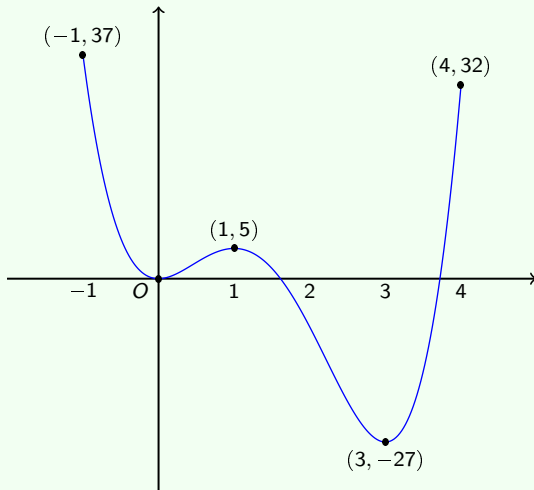
An absolute maximum or minimum is sometimes called a **global maximum or minimum**. The maximum and minimum values of f are called **extreme values** of f .

Definition 14: The number $f(c)$ is a

- **local maximum** value of f if $f(x) \leq f(c)$ when x near c .
- **local minimum** value of f if $f(x) \geq f(c)$ when x near c .



Example 32: The graph of the function $f(x) = 3x^4 - 16x^3 + 18x^2$, $(-1 \leq x \leq 4)$ is shown in the following figure.



You can see that $f(1) = 5$ is a local maximum, whereas the absolute maximum is $f(-1) = 37$. (This absolute maximum is not a local maximum because it occurs at an endpoint.) Also, $f(0) = 0$ is a local minimum and $f(3) = -27$ is both a local and an absolute minimum. Note that f has neither a local nor an absolute maximum at $x = 4$.

Theorem 8 (Extreme Value Theorem): *If f is continuous on a closed interval $[a, b]$, then f attains an absolute maximum value $f(c)$ and an absolute minimum value $f(d)$ at some numbers c and d in $[a, b]$.*

Theorem 9 (Fermat's Theorem): *If f has a local maximum or minimum at c , and if $f'(c)$ exists, then $f'(c) = 0$.*

Definition 15: A **critical number** of a function f is a number c in the domain of f such that either $f'(c) = 0$ or $f'(c)$ does not exist.

Example 33: Find the critical numbers of $f(x) = x^{3/5}(4-x)$.

Solution : The function is defined for all x . The Product Rule gives

$$\begin{aligned} f'(x) &= x^{3/5}(-1) + \frac{3}{5}x^{-2/5}(4-x) = \frac{-5x + 3(4-x)}{5x^{2/5}} \\ &= \frac{12-8x}{5x^{2/5}} \end{aligned}$$

Therefore $f'(x) = 0$ if $x = \frac{3}{2}$, and $f'(x)$ does not exist when $x = 0$. Thus the critical numbers are $\frac{3}{2}$ and 0.

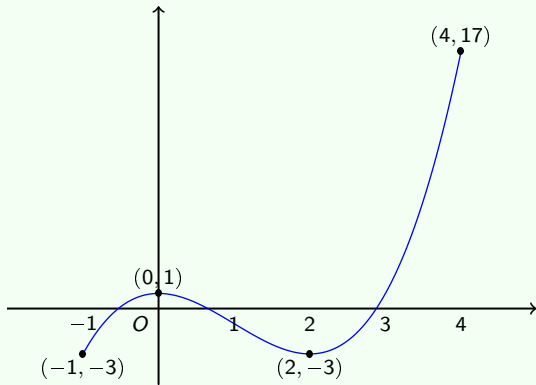
The Closed Interval Method

To find an absolute maximum or minimum of a continuous function f on a closed interval $[a, b]$, we note that either it is local or it occurs at an endpoint of the interval. Thus the following three-step procedure always works.

- Step 1: Find the values of f at the critical numbers of f in $[a, b]$.
- Step 2: Find the values of f at the endpoints of the interval.
- Step 3: The largest of the values from Step 1 and Step 2 is the absolute maximum value; the smallest of these values is the absolute minimum value.

Example 34: Find the absolute maximum and minimum values of the function $f(x) = x^3 - 3x^2 + 1$ on $[-1, 4]$.

Solution : Since f is continuous on $[-1, 4]$, we can use the Closed Interval Method: $f'(x) = 3x^2 - 6x = 3x(x - 2)$. Since $f'(x)$ exists for all x , the only critical numbers of f occur when $f'(x) = 0$, that is, $x = 0$ or $x = 2$. Notice that each of these critical numbers lies in the interval $(-1, 4)$. The values of f at these critical numbers are: $f(0) = 1$, $f(2) = -3$. The values of f at the endpoints of the interval are: $f(-1) = -3$, $f(4) = 17$. Comparing these four numbers, we see that the absolute maximum value is $f(4) = 17$ and the absolute minimum values are $f(2) = -3$ and $f(-1) = -3$.



OPTIMIZATION PROBLEMS

Steps in Solving Optimization Problems

- STEP 1: The first step is to read the problem carefully until it is clearly understood. Ask yourself: What is the unknown? What are the given quantities? What are the given conditions?
- STEP 2: In most problems it is useful to draw a diagram and identify the given and required quantities on the diagram.

Steps in Solving Optimization Problems

- STEP 3: Assign a symbol to the quantity that is to be maximized or minimized (let's call it Q for now). Also select symbols $a, b, c, \dots, x, y$ for other unknown quantities and label the diagram with these symbols. It may help to use initials as suggestive symbols, for example, A for area, h for height, t for time.
- STEP 4: Express Q in terms of some of the other symbols from Step 3.

Steps in Solving Optimization Problems

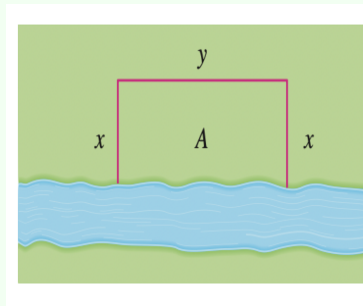
- STEP 5: If Q has been expressed as a function of more than one variable in Step 4, use the given information to find relationships (in the form of equations) among these variables. Then use these equations to eliminate all but one of the variables in the expression for Q . Thus Q will be expressed as a function of one variable x , say, $Q = f(x)$. Write the domain of this function.
- STEP 6: Use the knowing methods to find the absolute maximum or minimum value of f . In particular, if the domain of f is a closed interval, then the Closed Interval Method can be used.

Example 35: A farmer has 2400 ft of fencing and wants to fence off a rectangular field that borders a straight river. He needs no fence along the river. What are the dimensions of the field that has the largest area?

Solution :

The figure illustrates the general case. We wish to maximize the area A of the rectangle. Let x and y be the depth and width of the rectangle (in feet). Then we express A in terms of x and y :

$$A = xy$$



We want to express A as a function of just one variable, so we eliminate y by expressing it in terms of x . To do this we use the given information that the total length of the fencing is 2400 ft. Thus $2x + y = 2400$. From this equation we have $y = 2400 - 2x$, which gives

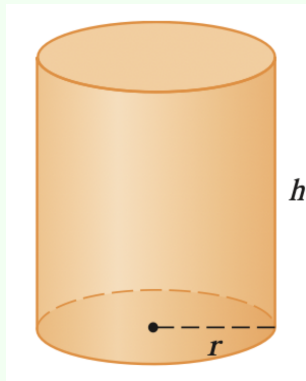
$$A = x(2400 - 2x) = 2400x - 2x^2, \quad (0 \leq x \leq 1200)$$

The derivative is $A'(x) = 2400 - 4x$, so to find the critical numbers we solve the equation $2400 - 4x = 0$, which gives $x = 600$. The maximum value of A must occur either at this critical number or at an endpoint of the interval. Since $A(0) = 0$, $A(600) = 720,000$, and $A(1200) = 0$, the Closed Interval Method gives the maximum value as $A(600) = 720,000$. Thus the rectangular field should be 600 ft deep and 1200 ft wide.

Example 36: A cylindrical can is to be made to hold 1 L of oil. Find the dimensions that will minimize the cost of the metal to manufacture the can.

Solution :

Draw the diagram as in the figure, where r is the radius and h the height (both in centimeters). In order to minimize the cost of the metal, we minimize the total surface area of the cylinder (top, bottom, and sides).



We see that the sides are made from a rectangular sheet with dimensions $2\pi r$ and h . So the surface area is

$$A = 2\pi r^2 + 2\pi rh$$

To eliminate h we use the fact that the volume is given as $1L = 1000\text{cm}^3$. Thus $\pi r^2 h = 1000 \Rightarrow h = \frac{1000}{\pi r^2}$. Substitution of this into the expression for A gives

$$A = 2\pi r^2 + 2\pi r \left(\frac{1000}{\pi r^2} \right) = 2\pi r^2 + \frac{2000}{r}, \quad (r > 0)$$

We have

$$A'(r) = 4\pi r - \frac{2000}{r^2} = 0 \Rightarrow r = \sqrt[3]{\frac{500}{\pi}}$$

Since the domain of A is $(0, \infty)$, we cannot use the argument of the previous example concerning endpoints. But we can observe that $A'(r) < 0$ for $r < \sqrt[3]{\frac{500}{\pi}}$ and $A'(r) > 0$ for $r > \sqrt[3]{\frac{500}{\pi}}$, so A is decreasing for all r to the left of the critical number and increasing for all r to the right. Thus $r = \sqrt[3]{\frac{500}{\pi}}$ must give rise to an absolute minimum. The corresponding value of h is

$$h = \frac{100}{\pi r^2} = \frac{1000}{\pi(500/\pi)^{2/3}} = 2\sqrt[3]{\frac{500}{\pi}} = 2r$$

Thus, to minimize the cost of the can, the radius should be $\sqrt[3]{\frac{500}{\pi}}$ cm and the height should be equal to twice the radius, namely, the diameter.

Theorem 10 (First Derivative Test for Absolute Extreme Values): Suppose that c is a critical number of a continuous function f defined on an interval.

- (a) If $f'(x) > 0$ for all $x < c$ and $f'(x) < 0$ for all $x > c$, then $f(c)$ is the absolute maximum value of f .
- (b) If $f'(x) < 0$ for all $x < c$ and $f'(x) > 0$ for all $x > c$, then $f(c)$ is the absolute minimum value of f .

Example 37: Find the point on the parabola $y^2 = 2x$ that is closest to the point $(1, 4)$.

Solution : The distance between the point $(1, 4)$ and the point (x, y) is

$$d = \sqrt{(x - 1)^2 + (y - 4)^2}$$

But if (x, y) lies on the parabola, then $x = \frac{1}{2}y^2$, so the expression for d becomes

$$d = \sqrt{\left(\frac{1}{2}y^2 - 1\right)^2 + (y - 4)^2}$$

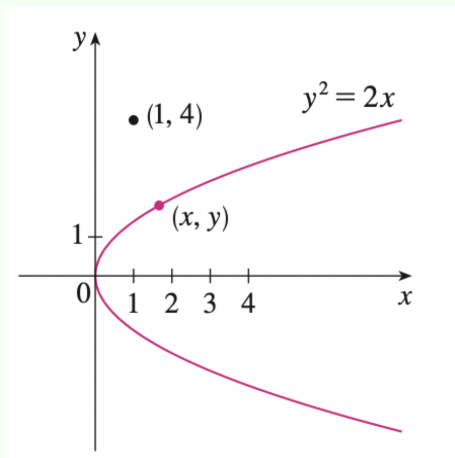
Instead of minimizing d , we minimize its square:

$$d^2 = f(y) = \left(\frac{1}{2}y^2 - 1\right)^2 + (y - 4)^2$$

Differentiating, we obtain

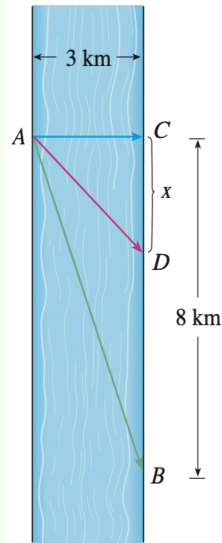
$$f'(y) = 2\left(\frac{1}{2}y^2 - 1\right)y + 2(y - 4) = y^3 - 8$$

so $f'(y) = 0$ when $y = 2$. Observe that $f'(y) < 0$ when $y < 2$ and $f'(y) > 0$ when $y > 2$, so by the First Derivative Test for Absolute Extreme Values, the absolute minimum occurs when $y = 2$. The corresponding value of x is $x = \frac{1}{2}y^2 = 2$. Thus the point on $y^2 = 2x$ closest to $(1, 4)$ is $(2, 2)$.



Example 38:

A man launches his boat from point A on a bank of a straight river, 3 km wide, and wants to reach point B , 8 km downstream on the opposite bank, as quickly as possible. He could row his boat directly across the river to point C and then run to B , or he could row directly to B , or he could row to some point D between C and B and then run to B . If he can row 6 km/h and run 8 km/h, where should he land to reach B as soon as possible? (We assume that the speed of the water is negligible compared with the speed at which the man rows.)



Solution : If we let x be the distance from C to D , then the running distance is $|DB| = 8 - x$ and the Pythagorean Theorem gives the rowing distance as $|AD| = \sqrt{x^2 + 9}$. We use the equation: $\text{time} = \frac{\text{distance}}{\text{velocity}}$, then the rowing time is $\frac{\sqrt{x^2 + 9}}{6}$ and the running time is $\frac{8 - x}{8}$. So the total time T as a function of x is

$$T(x) = \frac{\sqrt{x^2 + 9}}{6} + \frac{8 - x}{8}$$

The domain of this function T is $[0, 8]$. Notice that if $x = 0$, he rows to C and if $x = 8$, he rows directly to B . The derivative of T is

$$T'(x) = \frac{x}{6\sqrt{x^2 + 9}} - \frac{1}{8} = 0 \Leftrightarrow x = \frac{9}{\sqrt{7}}$$

The only critical number is $x = \frac{9}{\sqrt{7}}$. To see whether the minimum occurs at this critical number or at an endpoint of the domain $[0, 8]$, we evaluate T at all three points:

$$T(0) = 1.5, \quad T\left(\frac{9}{\sqrt{7}}\right) = 1 + \frac{\sqrt{7}}{8} \approx 1.33, \quad T(8) = \frac{\sqrt{73}}{6} \approx 1.42$$

Since the smallest of these values of T occurs when $x = \frac{9}{\sqrt{7}}$, the absolute minimum value of T must occur there. Thus the man should land the boat at a point $\frac{9}{\sqrt{7}}$ km (≈ 3.4 km) downstream from his starting point.

DERIVATIVES AND THE SHAPE OF A GRAPH

Theorem 11 (Increasing/Decreasing Test):

- (a) *If $f'(x) > 0$ on an interval, then f is increasing on that interval.*
- (b) *If $f'(x) < 0$ on an interval, then f is decreasing on that interval.*

Example 39: Find where the function $f(x) = 3x^4 - 4x^3 - 12x^2 + 5$ is increasing and where it is decreasing.

Solution : We have

$$f'(x) = 12x^3 - 12x^2 - 24x = 12x(x - 2)(x + 1)$$

To use the I/D Test we have to know where $f'(x) > 0$ and where $f'(x) < 0$. We have the following table

Interval	$12x$	$x - 2$	$x + 1$	$f'(x)$	$f(x)$
$x < -1$	-	-	-	-	decreasing
$-1 < x < 0$	-	-	+	+	increasing
$0 < x < 2$	+	-	+	-	decreasing
$2 < x$	+	+	+	+	increasing

Thus $f(x)$ is increasing on $(-1, 0)$ and $(2, +\infty)$, and is decreasing on $(-\infty, -1)$ and $(0, 2)$.

Recall from the previous sections that if f has a local maximum or minimum at c , then c must be a critical number of f (by Fermat's Theorem), but not every critical number gives rise to a maximum or a minimum. We therefore need a test that will tell us whether or not f has a local maximum or minimum at a critical number.

Theorem 12 (The First Derivative Test): *Suppose that c is a critical number of a continuous function f .*

- (a) *If f' changes from positive to negative at c , then f has a local maximum at c .*
- (b) *If f' changes from negative to positive at c , then f has a local minimum at c .*
- (c) *If f' does not change sign at c , then f has no local maximum or minimum at c .*

Example 40: Find the local maximum and minimum values of the function

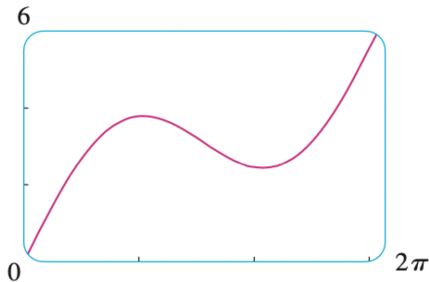
$$f(x) = x + 2 \sin x, \quad 0 \leq x \leq 2\pi$$

Solution : We have $f'(x) = 1 + 2 \cos x = 0$ when $\cos x = -\frac{1}{2} \Rightarrow x = \frac{2\pi}{3}$ and $x = \frac{4\pi}{3}$. We obtain the following table

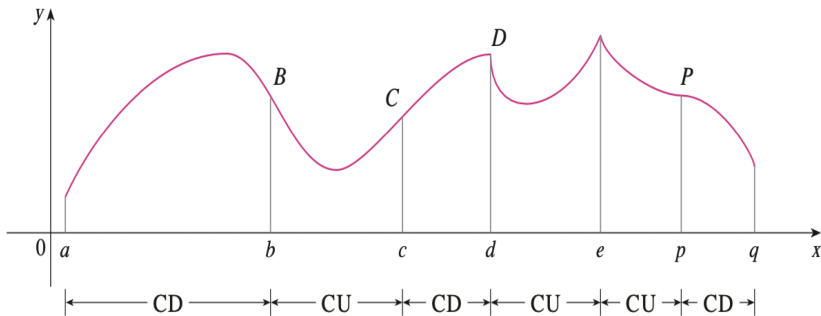
x	0	$\frac{2\pi}{3}$	$\frac{4\pi}{3}$	2π
$f'(x)$	+	0	-	0
$f(x)$	$\nearrow$	max	$\searrow$	min

The local maximum value of f is $f\left(\frac{2\pi}{3}\right) = \frac{2\pi}{3} + \sqrt{3}$.

The local minimum value of f is $f\left(\frac{4\pi}{3}\right) = \frac{4\pi}{3} - \sqrt{3}$.



Definition 16: If the graph of f lies above all of its tangents on an interval I , then it is called **concave upward** on I . If the graph of f lies below all of its tangents on I , it is called **concave downward** on I .



Theorem 13 (Concavity Test):

- (a) If $f''(x) > 0$ for all x in I , then the graph of f is concave upward on I .
- (b) If $f''(x) < 0$ for all x in I , then the graph of f is concave downward on I .

Definition 17: A point P on a curve $y = f(x)$ is called an **inflection point** if f is continuous there and the curve changes from concave upward to concave downward or from concave downward to concave upward at P .

Theorem 14 (The Second Derivative Test): Suppose f'' is continuous near c .

- (a) If $f'(c) = 0$ and $f''(c) < 0$, then f has a local maximum at c .
- (b) If $f'(c) = 0$ and $f''(c) > 0$, then f has a local minimum at c .

Example 41: Discuss the curve $y = x^4 - 4x^3$ with respect to concavity, points of inflection, and local maxima and minima.

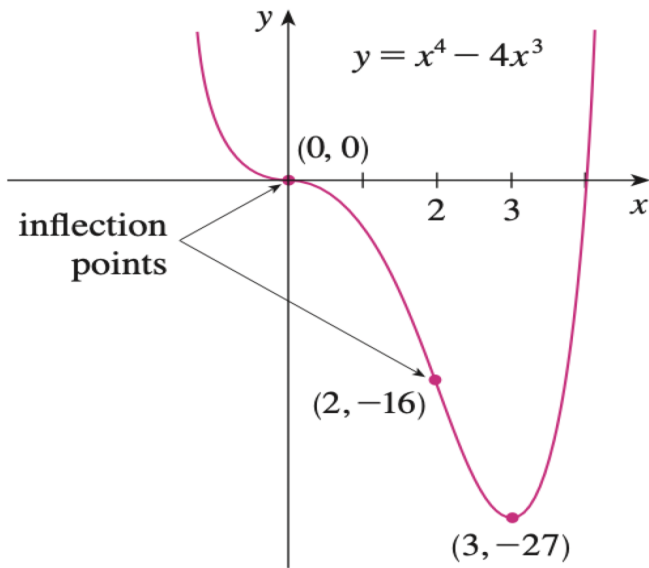
Solution : We have

$$f'(x) = 4x^3 - 12x^2 = 4x^2(x - 3)$$

$$f''(x) = 12x^2 - 24x = 12x(x - 2)$$

The critical numbers of f are $x = 0$ and $x = 3$, and $f''(0) = 0$, $f''(3) = 36 > 0$. Since $f'(3) = 0$ and $f''(3) > 0 \Rightarrow f(3) = -27$ is a local minimum. Since $f''(0) = 0$, the Second Derivative Test gives no information about the critical number 0. But since $f'(x) < 0$ for $x < 0$ and also for $0 < x < 3$, the First Derivative Test tells us that f does not have a local maximum or minimum at 0.

We see that $f''(x) > 0$ for all x in $(-\infty, 0)$ and $(2, +\infty)$, and $f''(x) < 0$ for x in $(0, 2)$. This means that the curve is concave upward on $(-\infty, 0)$ and $(2, +\infty)$, and is concave downward on $(0, 2)$. From this, the point $(0, 0)$ and $(2, -16)$ are the inflection points.



GUIDELINE FOR SKETCHING A CURVE

The following checklist is intended as a guide to sketching a curve $y = f(x)$ by hand. Not every item is relevant to every function. (For instance, a given curve might not have an asymptote or possess symmetry.) But the guidelines provide all the information you need to make a sketch that displays the most important aspects of the function.

A. Domain: It is often useful to start by determining the domain D of f , that is, the set of values of x for which $f(x)$ is defined.

B. Intercepts: The y -intercept is $f(0)$ and this tells us where the curve intersects the y -axis. To find the x -intercepts, we set $y = 0$ and solve for x .

C. Symmetry:

- (a) If $f(-x) = f(x)$ for all x in D , then f is an even function and the curve is symmetric about the y -axis. This means that our work is cut in half. If we know what the curve looks like for $x \leq 0$, then we need only reflect about the y -axis to obtain the complete curve.
- (b) If $f(-x) = -f(x)$ for all x in D , then f is an odd function and the curve is symmetric about the origin. Again we can obtain the complete curve if we know what it looks like for $x \leq 0$. [Rotate 180° about the origin].

- (c) If f is a periodic function with the period T and we know what the graph looks like in an interval of length T , then we can use translation to sketch the entire graph.

D. Asymptotes:

- (a) Horizontal Asymptotes: If either $\lim_{x \rightarrow \pm\infty} f(x) = L$, then the line $y = L$ is a horizontal asymptotes of the curve $y = f(x)$.
- (b) Vertical Asymptotes: The line $x = a$ is a vertical asymptote if at least one of the following statements is true

$$\lim_{x \rightarrow a} f(x) = \pm\infty, \quad \lim_{x \rightarrow a^+} f(x) = \pm\infty, \quad \lim_{x \rightarrow a^-} f(x) = \pm\infty$$

(c) Slant Asymptotes: The line $y = mx + b$ is called a slant asymptote of the curve $y = f(x)$ if

$$\lim_{x \rightarrow \pm\infty} [f(x) - (mx + b)] = 0$$

We can determine the values of m and b by the following formulas (if the limits exist)

$$m = \lim_{x \rightarrow \infty} \frac{f(x)}{x}, \quad b = \lim_{x \rightarrow \infty} [f(x) - mx]$$

E. Intervals of Increase or Decrease: Compute $f'(x)$ and find the intervals on which $f'(x)$ is positive (f is increasing) and the intervals on which $f'(x)$ is negative (f is decreasing).

F. Local Maximum and Minimum Values: Find the critical numbers of f . Then use the First Derivative Test or the Second Derivative Test.

G. Concavity and Points of Inflection: Compute $f''(x)$ and use the Concavity Test. The curve is concave upward where $f''(x) > 0$ and concave downward where $f''(x) < 0$. Inflection points occur where the direction of concavity changes.

H. Sketch the Curve: Use the information in items A-G, draw the graph.

Example 42: Sketch the curve $y = \sqrt[3]{x(x-2)^2}$.

Solution : This function defines for all x . The y -intercept is $(0,0)$ and the x -intercept is $(0,0)$ and $(2,0)$. The function is neither even nor odd.

The curve does not have vertical and horizontal asymptotes.

To find the slant asymptote we consider the following limits:

$$m = \lim_{x \rightarrow \infty} \frac{f(x)}{x} = \lim_{x \rightarrow \infty} \frac{\sqrt[3]{x(x-2)^2}}{x} = \lim_{x \rightarrow \infty} \sqrt[3]{\left(1 - \frac{2}{x}\right)^2} = 1$$

$$\begin{aligned} b &= \lim_{x \rightarrow \infty} [f(x) - mx] = \lim_{x \rightarrow \infty} [\sqrt[3]{x(x-2)^2} - x] \\ &= \lim_{x \rightarrow \infty} \frac{x(x-2)^2 - x^3}{\sqrt[3]{x^2(x-2)^4} + x\sqrt[3]{x(x-2)^2} + x^2} = -\frac{4}{3} \end{aligned}$$

The slant asymptote is $y = x - \frac{4}{3}$.

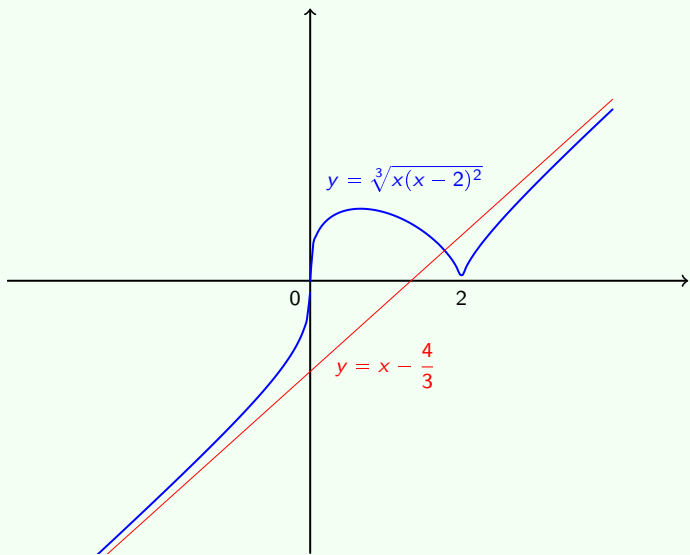
We have $f'(x) = \frac{3x-2}{3\sqrt[3]{x^2(x-2)}}$. The critical numbers of f are $x = \frac{2}{3}$ ($f'(x) = 0$), $x = 0$, $x = 2$ ($f'(x)$ does not exist).

We have the following table

x	$-\infty$		0		$2/3$		2		$+\infty$
$f'(x)$		+		+	0	-		+	$+\infty$
$f(x)$		$\nearrow$	0	$\nearrow$	$\frac{2\sqrt[3]{4}}{3}$	$\searrow$	0	$\nearrow$	$+\infty$

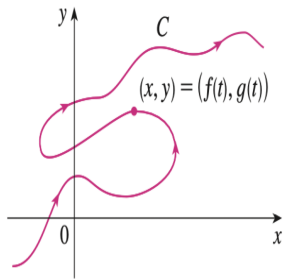
From the above table, f has a local maximum at $\frac{2}{3}$ and a local minimum at 2 .

We have $f''(x) = -\frac{2}{9} \cdot \frac{x^2 - 2x + 4}{\sqrt[3]{x^5(x-2)^4}}$. If $x < 0$ then $f''(x) > 0$, the curve is concave upward. If $x > 0$ then $f''(x) < 0$, the curve is concave downward.



CURVE DEFINED BY PARAMETRIC EQUATIONS

Imagine that a particle moves along the curve C shown in the figure. It is impossible to describe C by an equation of the form $y = f(x)$ because C fails the Vertical Line Test. But the x - and y -coordinates of the particle are functions of time t and so we can write $x = f(t)$ and $y = g(t)$.



Such a pair of equations is often a convenient way of describing a curve and gives rise to the following definition.

Definition 18: Suppose that x and y are both given as functions of a third variable t (called a **parameter**) by the equations

$$x = f(t), \quad y = g(t)$$

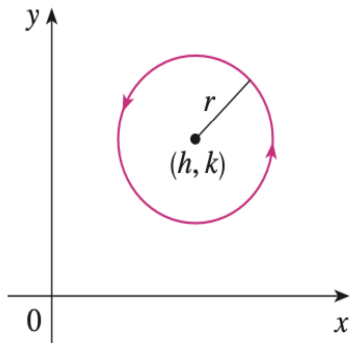
(called **parametric equations**). Each value of t determines a point (x, y) , which we can plot in a coordinate plane. As t varies, the point $(x, y) = (f(t), g(t))$ varies and traces out a curve C , which we call a **parametric curve**.

In some cases, we also consider the parametric equations $x = f(t)$, $y = g(t)$ where t lies in a finite interval $[a, b]$. Then the curve has **initial point** $(f(a), g(a))$ and **terminal point** $(f(b), g(b))$.

The Circle

The circle with center (h, k) and radius r is given by

$$x = h + r \cos t, \quad y = k + r \sin t, \quad 0 \leq t \leq 2\pi$$

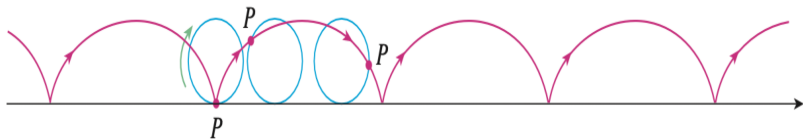


The Cycloid

The curve traced out by a point P on the circumference of a circle as the circle rolls along a straight line is called a cycloid. If the circle has radius r and rolls along the x -axis, then the parametric equations for the cycloid are

$$x = r(t - \sin t), \quad y = r(1 - \cos t) \quad t \in \mathbb{R}$$

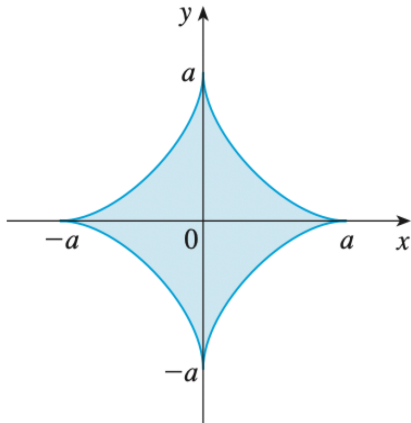
One arch of the cycloid comes from one rotation of the circle and so is described by $0 \leq t \leq 2\pi$.



The Astroid

The astroid is a curve given by

$$x = a \sin^3 t, \quad y = a \cos^3 t \quad 0 \leq t \leq 2\pi$$



If we want to find a derivative at a point on the curve given by parametric equations $x = f(t)$, $y = g(t)$, we can use the Chain Rule:

$$\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$$

Denote $\frac{dy}{dx} = y'_x$, $\frac{dy}{dt} = y'_t$, $\frac{dx}{dt} = x'_t$ and if $x'_t \neq 0$, we have

$$y'_x = \frac{y'_t}{x'_t}$$

and for the second derivative

$$y''_{xx} = \frac{x'_t y''_{tt} - y'_t x''_{tt}}{(x'_t)^3}$$

Example 43: Find y'_x and y''_{xx} if the curve is given by

$$x(t) = t^2 - 2t, \quad y(t) = t^3 + 1$$

Solution : We have

$$x'(t) = 2t - 2, \quad x''(t) = 2, \quad y'(t) = 3t^2, \quad y''(t) = 6t$$

Therefore

$$\begin{aligned} y'_x &= \frac{y'_t}{x'_t} = \frac{3t^2}{2t - 2} = \frac{3t^2}{2(t - 1)} \\ y''_{xx} &= \frac{x'_t y''_{tt} - y'_t x''_{tt}}{(x'_t)^3} = \frac{(2t - 2) \cdot 6t - 3t^2 \cdot 2}{(2t - 2)^3} = \frac{3t(t - 2)}{4(t - 1)^3} \end{aligned}$$

Example 44: Find the tangent line to the cycloid

$$x = 2(t - \sin t), y = 2(1 - \cos t)$$

at the point when $t = \frac{\pi}{3}$.

Solution : We have

$$\frac{dy}{dx} = y'_x = \frac{y'_t}{x'_t} = \frac{\sin t}{1 - \cos t} \Rightarrow \frac{dy}{dx} \Big|_{t=\pi/3} = \frac{\sqrt{3}/2}{1 - 1/2} = \sqrt{3}$$

When $t = \frac{\pi}{3}$, then $x = \frac{2\pi}{3} - \sqrt{3}$, $y = 1$. We have

$$y - 1 = \sqrt{3} \left(x - \frac{2\pi}{3} + \sqrt{3} \right) \Leftrightarrow y = \sqrt{3}x - 2\pi\sqrt{3} + 4$$

POLAR COORDINATES

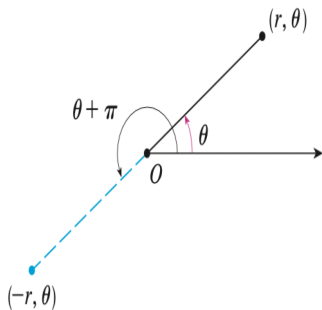
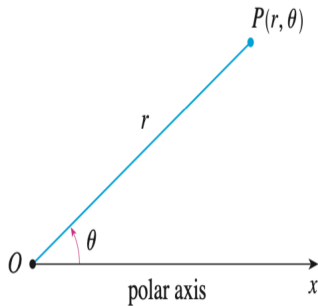
A coordinate system represents a point in the plane by an ordered pair of numbers called coordinates. Usually we use Cartesian coordinates, which are directed distances from two perpendicular axes. Now we introduce the **polar coordinate system**, which is more convenient for many purposes.

We choose a point in the plane that is called the **pole** (or **origin**) and is labeled O . Then we draw a half-line starting at O called the **polar axis**. This axis is usually drawn horizontally to the right and corresponds to the positive x -axis in Cartesian coordinates.

If P is any other point in the plane, let r be the distance from O to P and let θ be the angle (usually measured in radians) between the polar axis and the line OP . Then the point P is represented by the ordered pair (r, θ) and r, θ are called **polar coordinates** of P .

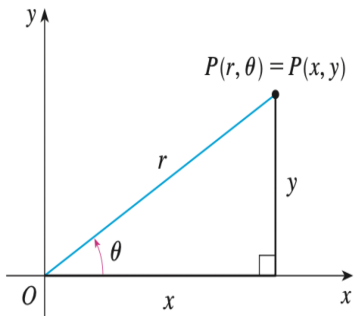
We use the convention that an angle is positive if measured in the counterclockwise direction from the polar axis and negative in the clockwise direction. If $P = O$, then $r = 0$ and we agree that $(0, \theta)$ represents the pole for any value of θ .

We extend the meaning of polar coordinates (r, θ) to the case in which r is negative by agreeing that the points (r, θ) and $(-r, \theta)$ lie on the same line through O and at the same distance $|r|$ from O , but on opposite sides of O .



If $r > 0$, the point (r, θ) lies in the same quadrant as θ ; if $r < 0$, it lies in the quadrant on the opposite side of the pole. Notice that $(-r, \theta)$ represents the same point as $(r, \theta + \pi)$. The connection between polar and Cartesian coordinates can be seen from the figure (in the next slide), in which the pole corresponds to the origin and the polar axis coincides with the positive x -axis. If the point P has Cartesian coordinates (x, y) and polar coordinates (r, θ) , then, from the figure, we have

$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases} \Leftrightarrow \begin{cases} r = \sqrt{x^2 + y^2} \\ \tan \theta = \frac{y}{x} \end{cases}$$



The Polar Curve: The graph of a polar equation $r = f(\theta)$, or more generally $F(r, \theta) = 0$, consists of all points P that have at least one polar representation (r, θ) whose coordinates satisfy the equation.

Example 45: Here are some curves in polar coordinates:

(1) $r = 1$ is a circle with center at origin and radius 1:

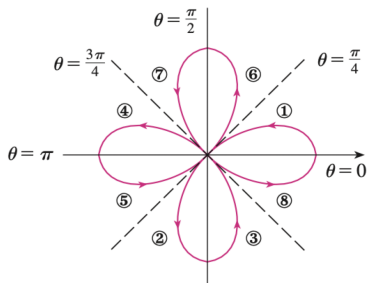
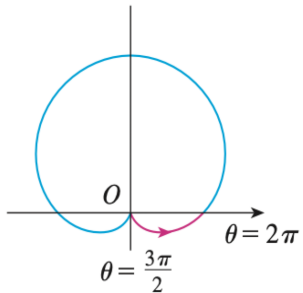
$$x^2 + y^2 = 1$$

(2) $r = 2 \cos \theta$ is a circle with center at $(1, 0)$ and radius 1:

$$(x - 1)^2 + y^2 = 1 \text{ or } x^2 + y^2 = 2x$$

(3) $r = 1 + \sin \theta$ is a cardioid.

(4) $r = \cos 2\theta$ is a four-leaved rose.



— THE END —